

Adverse Drug Reactions :-

- According to WHO, ADR is defined as "A response to a drug which is noxious and unintended and which occurs at doses normally used in man for the prophylaxis, diagnosis or therapy of diseases."
- Adverse event is defined as "Any medical occurrence temporally associated with the use of a medicinal product but not necessarily causally related."

⇒ Types of ADR :-

→ Type A - Augmented Pharmacologic effects

- Adverse effect that are known to occur from the pharmacology of the drug and are dose-related.

→ Type B (Bizarre effects)

- Adverse effects that occur ~~up~~ unpredictably and often have a high rate of morbidity and mortality.

→ Type C [Chronic effects]

- Adverse effects that only occur during prolonged treatment and not with single doses.

→ Type D [delayed effects]

- Adverse effects that occur sometime from treatment; either in the children of treated patient.

→ Type E [end of treatment effects]

- Adverse effects that occur when a drug is stopped especially when it is stopped suddenly.

⇒ Detection methods of ADRs :-

① pre-marketing studies -

- During the development of new medicines, their safety is tested in animal models.
- Specific animal studies for carcinogenicity, teratogenicity.

② Post marketing surveillance -

- pharmacovigilance methods are used for detection of risk and for the collection of risk information.
- Health Care practitioners should see it as a part of professional duty report ADR result in a patient under his care.

Two methods are commonly used :-

① Cohort studies

② Case-control studies

③ Communicating ADRs :-

- Knowledge about rational and safe use of medicines needs to be provided.

① During basic training of health professionals

② By specially designated drug information centers

④ Postal survey method -

- This consists of a specific drug related questions
- This method is primarily used for ADR monitoring of a new drug. 1-2 yrs after the drug is launched.

⇒ Severity of ADRs

- Severity describes the extent to which the ADR influence the everyday life of a patient.

- ① mild severity :- No antidote or treatment is required (level 1 & 2)
- ② moderate :- A change in the treatment but no necessary discontinuation, specific treatment may be required. (level 3 & 4)
- ③ severe :- ADR may be life threatening and requires immediate discontinuation of drug (level 5, 6, 7)
- ④ lethal :- contributes to patient's death

⇒ Seriousness of ADRs

- It is related to its life threatening nature and is defined as any untoward reaction to the medicinal product that may result in death and requires patient hospitalization.

- Death, life-threatening
- Hospitalization, disability
- Birth defect.

⇒ Management of ADR

- The main and primary step in management is withdrawal of suspected drugs.
- monitoring patient who are at greater risk developing ADRs.
- Changing dose of drug.
- Replacement with alternate medicine.
- Stimulating health care professionals in reporting ADR.
- Obtaining feedback about the reported reactions.

Reporting of ADRs :-

What to report - All type of suspected ADRs - irrespective of whether they are known or unknown, serious or non-serious, frequent or rare.

Who should report - All health care and non-health care professionals, hospitals, health centres, consumers, pharmaceutical industry.

How and whom to report :-

- use the ADR reporting form which is available on official sites of IPC as well as CDSCO.

- Helpline Number - 1800-180-3024